

Bad Data → Bad Algorithms

Irrelevant Features, Feature Engineering & Overfitting in Machine Learning

1. Irrelevant Features

Garbage in → Garbage out

Irrelevant Features & The Reality

Useful Features

When predicting house prices, features like area, bedrooms, location, and age provide critical patterns for the model.

Irrelevant Features

Adding owner_name, favorite_color, or house_id introduces noise. More information does **not** always mean a better model!

2. Feature Engineering



1. Feature Selection

Choosing the most useful features from existing data. E.g., filtering 100 features down to the top 15 useful ones.



2. Feature Extraction

Combining features to build smarter ones. E.g., merging height and weight to compute BMI (related to dimensionality reduction).



3. Creating New Features

Gathering fresh external data. E.g., adding `distance_to_city_center` when predicting house prices.

3. Overfitting 🔥

Human vs. Machine Overgeneralization

What is Overfitting?

99%

Training vs 65% Test Accuracy

The Taxi Driver Analogy

You visit a new country, a single taxi driver cheats you, and you conclude: *"All drivers in this country are dishonest."*

In ML: The model learns the training data and noise by heart, but fails completely on new, unseen data. It knows the past, but doesn't recognize the future!

The "Country Name" Trap (Noise vs Pattern)

- ⚠️ **Life Satisfaction Prediction:** Passing Country name as a feature to a complex model.
- ✖️ **Spurious Correlation:** New Zealand (7.3), Norway (7.6), Sweden (7.3), Switzerland (7.5) all contain the letter "w".
- 💡 **False Rule Learned:** *"If country name has 'w' $\rightarrow$ life satisfaction > 7 ".*
- 💀 **Failure on New Data:** Rwanda or Zimbabwe will completely break this absurd rule because it learned random noise, not true laws!

Why Does Overfitting Happen?

The Recipe for Overfitting

**Small Dataset + Complex Model + Noisy Data =
OVERFITTING**

The Core Issue

The model lacks sufficient data to extract the genuine underlying pattern, forcing it to memorize random fluctuations and anomalies instead.

How to Fight Overfitting?



1. Simplify the Model

Switch from high-degree polynomial to linear models, or decrease the number of features used.



2. More Training Data

Scale up from 1,000 to 100,000 samples. More data makes it easier to separate true patterns from noise.



3. Reduce Noise

Clean up the dataset, correct erroneous entries, and handle outliers appropriately.

9. Regularization

What is Regularization?

Constraining a model to prevent it from becoming overly complex and flexible.

For example, in a linear equation $y = \theta_0 + \theta_1 x$, if we force parameters like θ_1 to stay small or zero, we restrict extreme curvature and maintain balance.

The Spectrum of Models

- **Underfitting:** Too simple (Bad training & test performance)
- **Good Generalization:** Right balance (Good training & test performance)
- **Overfitting:** Too complex (Great training, terrible test performance)

The Ultimate Pipeline

- ✓ **Irrelevant Feature:** A feature providing no useful information or creating noise.
- ✓ **Feature Selection / Extraction:** Filtering or constructing smarter variables.
- ✓ **Overfitting:** Memorizing training noise rather than generalizing.
- ✓ **Regularization:** Penalizing model complexity to enhance test accuracy.
- 🔥 **The Golden Rule:** Bad Data → Bad Features → Bad Model → Poor Generalization.

Any Questions?

Ready to build better, generalized machine learning models!

Machine Learning Foundations | Practice & Theory